

Visualizing Uncertainty

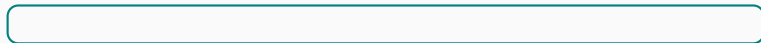
How to Show What the Data Doesn't Know

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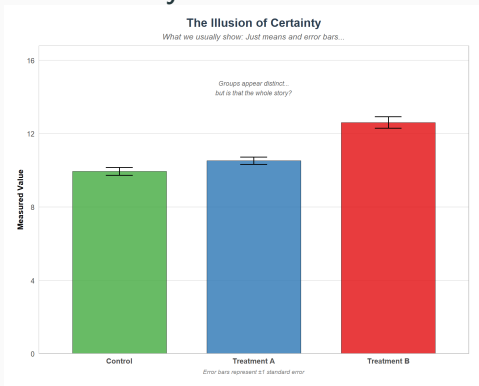
March 31, 2026

UNILORIN R Users Group



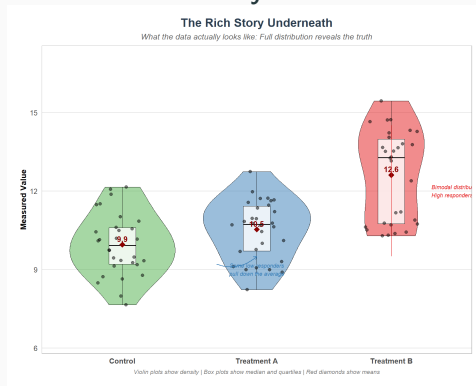
The Illusion of Certainty

What we usually show



"Just the means and error bars..."

What the data actually looks like



"...hides the rich story underneath"

Myth: Averages represent the data

Reality: Averages hide the story

Today's Journey

 The problem with summaries

 Basic uncertainty tools

 Modern solutions

 Case study in R

 Best practices

	Takeaways
	Live coding
	Modern methods
	Basic tools
	Foundations

Setup: Load Our Tools

```
# Load required packages
library(tidyverse)    # For ggplot2 and data manipulation
library(ggdist)       # For distribution visualizations
library(raincloudplots) # For raincloud plots
library(patchwork)    # For combining plots
library(gapminder)    # Example dataset

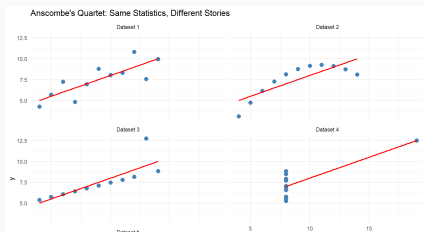
# Set a clean theme for all plots
theme_set(theme_minimal(base_size = 14) +
           theme(plot.title = element_text(face = "bold"),
                 legend.position = "bottom"))

# Create data directory
if(!dir.exists("figures")) dir.create("figures")
```

Packages to install beforehand

```
install.packages(c("tidyverse", "ggdist", "raincloudplots", "patchwork",
                  "gapminder"))
```

The Classic Warning: Anscombe's Quartet



	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Mean x	9.0	9.0	9.0	9.0
Mean y	7.5	7.5	7.5	7.5
Variance x	11.0	11.0	11.0	11.0
Variance y	4.12	4.12	4.12	4.12
Correlation	0.816	0.816	0.816	0.816
Regression	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$

*"Summary statistics are not sufficient. **Always visualize your data.**"*

Creating Our Example Dataset

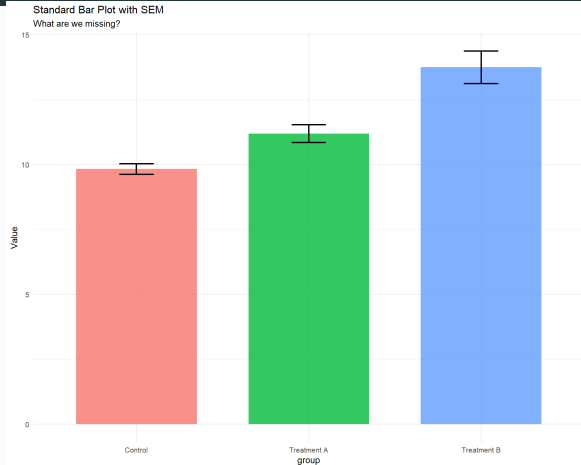
```
# Generate realistic-looking but fake data
set.seed(2024) # For reproducibility

# Create three groups with different distributions
n <- 100
example_data <- tibble(
  group = rep(c("Control", "Treatment A", "Treatment B"), each = n),
  value = c(
    # Control: normal distribution
    rnorm(n, mean = 10, sd = 2),
    # Treatment A: bimodal distribution
    c(rnorm(n/2, mean = 8, sd = 1), rnorm(n/2, mean = 14, sd = 1.5)),
    # Treatment B: skewed distribution
    rlnorm(n, meanlog = log(12), sdlog = 0.5)
  )
)

# Quick peek at the data
head(example_data, 8)
```

The Traditional Approach: What's Wrong Here?

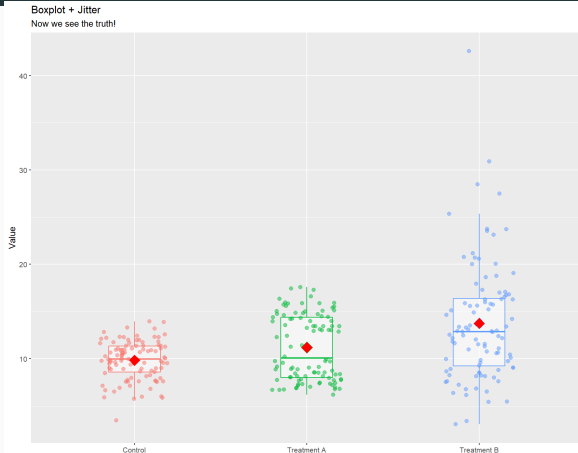
```
# The "standard" bar plot with error bars
p_bar <- example_data %>%
  group_by(group) %>%
  summarise(mean = mean(value),
            sd = sd(value),
            n = n(),
            se = sd / sqrt(n)) %>%
  ggplot(aes(x = group, y = mean, fill = group)) +
  geom_col(width = 0.7, alpha = 0.8) +
  geom_errorbar(aes(ymin = mean - se,
                  ymax = mean + se),
               width = 0.2, size = 1) +
  labs(title = "Standard Bar Plot with SEM",
       subtitle = "What are we missing?",
       y = "Value", x = "") +
  theme(legend.position = "none")
```



- ✗ Hides multimodality
- ✗ Hides outliers
- ✗ Hides sample size

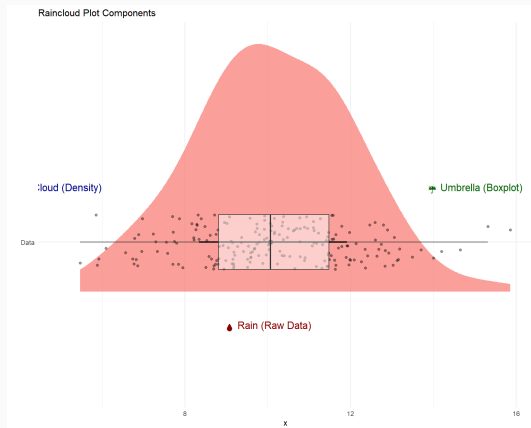
Step 1: Show the Raw Data

```
# Add jittered points to see distribution
p_jitter <- example_data %>%
  ggplot(aes(x = group, y = value, color = group)) +
  geom_boxplot(width = 0.3, alpha = 0.5,
               outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5,
              size = 2) +
  stat_summary(fun = mean, geom = "point",
               size = 5, shape = 23,
               fill = "red") +
  labs(title = "Boxplot + Jitter",
       subtitle = "Now we see the truth!",
       y = "Value", x = "") +
  theme(legend.position = "none")
```



- ✓ Shows multimodality in Treatment A
- ✓ Shows skewness in Treatment B
- ✓ Shows outliers

The Raincloud Plot: Best of Both Worlds



Rain
Raw data points 🌧️

Cloud
Density distribution ☁️

Umbrella
Boxplot/summary 🌂

Creating a Raincloud Plot in R

```
# Method 1: Using ggdist (more flexible)
p_raincloud <- example_data %>%
  ggplot(aes(x = group, y = value, fill = group)) +

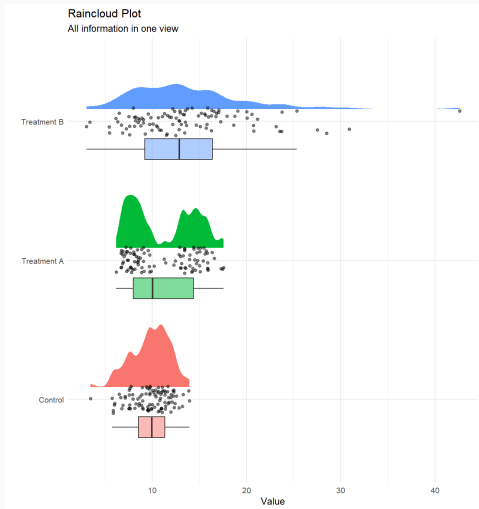
  # The "cloud" - half-eye (density)
  stat_halfeye(adjust = 0.5, width = 0.5,
               justification = -0.2,
               .width = 0, point_colour = NA) +

  # The "rain" - jittered points (using jitter parameters only)
  geom_jitter(width = 0.1, height = 0,
              alpha = 0.5, size = 1.5) +

  # The "umbrella" - boxplot
  geom_boxplot(width = 0.15, alpha = 0.5,
               outlier.shape = NA,
               position = position_nudge(x = -0.2)) +

  labs(title = "Raincloud Plot",
        subtitle = "All information in one view",
        y = "Value", x = "") +
  theme_minimal() +
  theme(legend.position = "none") +
  coord_flip()
```

The Raincloud Plot



What we now see clearly:

- Control: Nice normal distribution
- Treatment A: Bimodal! Some respond well, some don't
- Treatment B: Right-skewed with a long tail

Uncertainty in Relationships

```
# Using gapminder data
gapminder_2007 <- gapminder %>%
  filter(year == 2007)

# Linear model with confidence bands
p_regression <- gapminder_2007 %>%
  ggplot(aes(x = gdpPerCap,
             y = lifeExp)) +

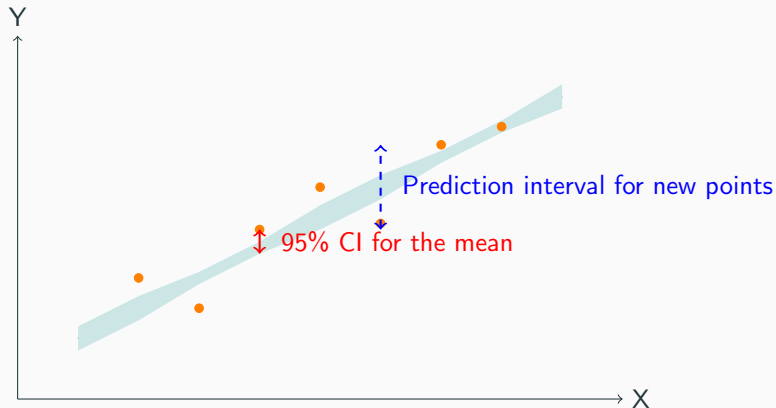
# Points with transparency
  geom_point(alpha = 0.5, size = 3,
            aes(color = continent)) +

# LOESS smooth with confidence band
  geom_smooth(method = "loess",
             se = TRUE,
             color = "darkblue",
             fill = "lightblue",
             alpha = 0.3) +

  scale_x_log10() +
  labs(title = "Life Expectancy vs GDP",
       subtitle = "The band shows our uncertainty",
       x = "GDP per Capita (log scale)",
       y = "Life Expectancy",
       color = "Continent")
```



What Does That Band Mean?



Two Types of Uncertainty

Confidence Band Where the *true line* probably lies

Prediction Band Where *new data points* will probably fall

LIVE CODING: Real-World Example

Dataset: Student Performance Factors

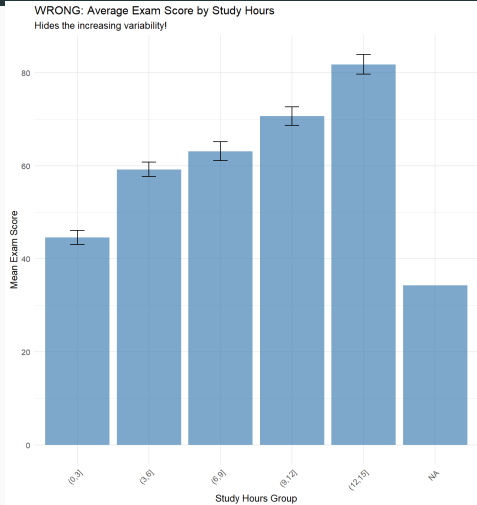
Let's analyze how study hours relate to exam scores, but with a twist - we'll visualize the uncertainty!

```
# Generate realistic student data
set.seed(2024)
student_data <- tibble(
  study_hours = round(runif(200, 0, 15), 1),
  base_score = 40 + 3*study_hours,
  exam_score = base_score +
    rnorm(200, mean = 0, sd = 10) +
    # Add heteroscedasticity (more variation with more study)
    rnorm(200, mean = 0, sd = study_hours * 0.8)
)

# Ensure scores are within 0-100 range
student_data <- student_data %>%
  mutate(exam_score = pmax(0, pmin(100, exam_score)))
```

LIVE: The Wrong Way (Don't Do This!)

```
# WRONG WAY - Hides increasing uncertainty
p_wrong <- student_data %>%
  mutate(hours_group = cut(study_hours,
                           breaks = seq(0, 15, by = 3))) %>%
  group_by(hours_group) %>%
  summarise(mean_score = mean(exam_score),
            se = sd(exam_score)/sqrt(n())) %>%
  ggplot(aes(x = hours_group, y = mean_score)) +
  geom_col(fill = "steelblue", alpha = 0.7) +
  geom_errorbar(aes(ymin = mean_score - se,
                   ymax = mean_score + se),
               width = 0.2) +
  labs(title = "WRONG: Average Exam Score by Study Hours",
       subtitle = "Hides the increasing variability!",
       x = "Study Hours Group", y = "Mean Exam Score") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



⚠ What's hidden? The more you study, the MORE variable your score becomes!

LIVE: The Right Way (Show the Spread!)

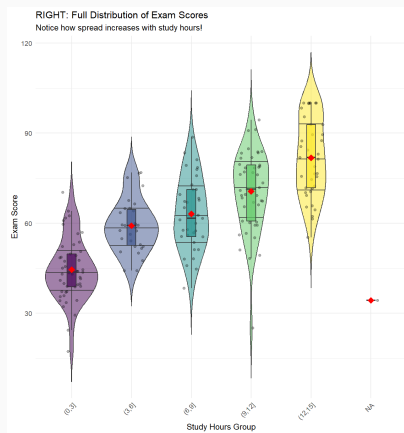
```
# RIGHT WAY - Shows increasing uncertainty
p_right <- student_data %>%
  mutate(hours_group = cut(study_hours,
                           breaks = seq(0, 15, by = 3))) %>%
  ggplot(aes(x = hours_group, y = exam_score,
            fill = hours_group)) +

  # Raincloud approach
  geom_violin(alpha = 0.5, trim = FALSE,
             draw_quantiles = c(0.25, 0.5, 0.75)) +
  geom_jitter(alpha = 0.3, width = 0.2, size = 1.5) +
  geom_boxplot(width = 0.15, alpha = 0.7,
              outlier.shape = NA) +

  # Add mean point
  stat_summary(fun = mean, geom = "point",
              size = 4, shape = 18, color = "red") +

  labs(title = "RIGHT: Full Distribution of Exam Scores",
       subtitle = "Notice how spread increases with study hours!",
       x = "Study Hours Group", y = "Exam Score") +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_viridis_d()
```

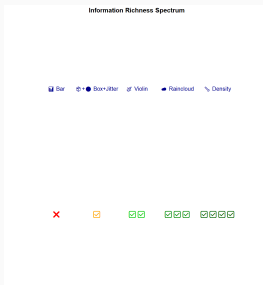

The Revelation: Heteroscedasticity



Critical Finding!

The variability in exam scores **increases** with study hours. Why? Because other factors (sleep, motivation, natural ability) start to matter more as hours increase.

Visualization Spectrum: From Simple to Rich



Simple Bar chart (least informative)

Basic Boxplot + jitter

Intermediate Violin plot

Advanced Raincloud plot

Complex Gradient density (most informative)

Best Practices for Uncertainty Visualization

Always show:

- ✓ Raw data points
- ✓ Distribution shape
- ✓ Sample size
- ✓ Central tendency
- ✓ Spread/variability

Avoid:

- ✗ Dynamite plots
- ✗ Only showing means
- ✗ Hiding outliers
- ✗ Misleading error bars

The Golden Rule: If your visualization doesn't show the distribution, it's hiding the truth.

Choosing Your Uncertainty Visualization

Sample Size	Question	Recommended Plot
 Small N		
$n < 30$	No trend	Jitter Plot + Summary Stats
	With trend	 Jitter Plot + Smoothing + CI
 Large N		
$n \geq 30$	No trend	 Raincloud Plot or Violin Plot
	With trend	 Raincloud Plot + LOESS/GAM Smoothing

Table 1: Decision guide for uncertainty visualization

 **Pro Tip:** Always show raw data when possible. If overplotting occurs, use transparency (alpha) or switch to density-based visualizations.

R Packages for Uncertainty Visualization

Core Tidyverse

- **ggplot2** - Foundation
- **dplyr** - Data manipulation
- **tidyr** - Data reshaping

Specialized

- **ggdist** - Distribution plots
- **raincloudplots** - Rainclouds
- **ggridges** - Joyplots
- **ggeffects** - Model predictions

Extensions

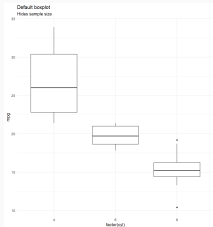
- **patchwork** - Combine plots
- **plotly** - Interactivity
- **gganimate** - Animation

Quick Install

```
install.packages(c("ggdist", "raincloudplots", "gggridges", "patchwork"))
```

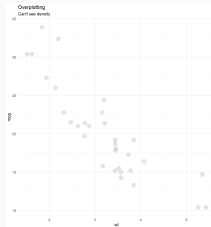
Mistakes I've Made (So You Don't Have To)

Mistake 1: Default Settings



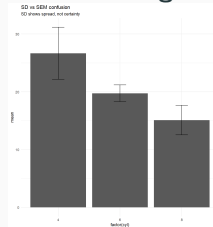
R's defaults hide uncertainty

Mistake 2: Too Much Transparency



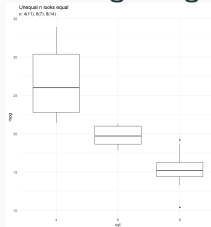
Overplotting obscures patterns

Mistake 3: Wrong Error Bars



Using SD vs. CI incorrectly

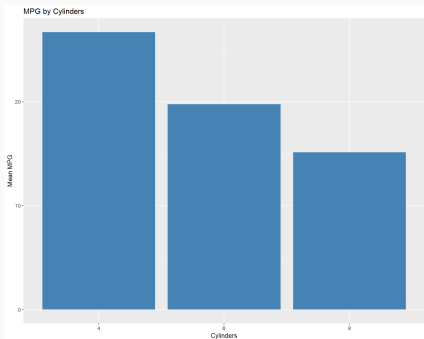
Mistake 4: Ignoring Sample Size



All groups look equally certain

Challenge: Improve This Plot

```
# Terrible plot - your job to fix it!  
data(mtcars)  
  
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +  
  geom_bar(stat = "summary", fun = "mean",  
    fill = "steelblue") +  
  labs(title = "MPG by Cylinders",  
    x = "Cylinders", y = "Mean MPG")
```



Your mission:

1. Show the actual distribution
2. Add raw data points
3. Indicate sample size
4. Show variability
5. Add a better title

Solution will be shared after the session!

Resources to Continue Your Journey

Books

- "Fundamentals of Data Visualization" - Claus Wilke
- "ggplot2: Elegant Graphics for Data Analysis" - Hadley Wickham
- "Statistical Rethinking" - Richard McElreath

Online

- From Data to Viz
- ggplot2 documentation
- ggdist documentation

Papers & Articles

- "Raincloud plots: a multi-platform tool for robust data visualization" - Allen et al. (2021)
- "Beyond Bar and Line Graphs" - Weissgerber et al. (2015)
- "Visualizing Uncertainty" - Hullman (2019)

Packages to explore

- **tidybayes** - Bayesian uncertainty
- **bayesplot** - MCMC visualizations
- **ggforce** - Advanced ggplot2

Thank You!

Questions?

<http://www.youtube.com/@olusanyaolubusoye9061>